ASIC Implementation of a **Pipelined** Energy-Efficient and Throughput-Optimized CNN **MAC Accelerator** for Edge AI Applications

- **Xcelium** → Simulation
- **Genus** → Synthesis
- **Innovus** → Physical Design
- **Tempus** → STA (if available)
- **Voltus** → Power (optional)
- **SimVision** → Waveform

RTL (Verilog)
↓
Functional Simulation (Xcelium)
↓
Synthesis (Genus)
↓
Gate-Level Simulation
↓
Floorplan (Innovus)
↓
Placement
↓
Clock Tree Synthesis
↓
Routing
↓
Post-Layout STA (Tempus)
↓
GDS-II Generation
↓
PPA Comparison

PHASE 1 — FPGA PROTOTYPE (Vivado)

Week	Stage	Work to Do	Output
1	CNN Hardware Basics	Study convolution operation, MAC role in CNN	Architecture document
2	Fixed-Point Study	Decide 8-bit or 16-bit fixed-point format	Data format document
3	Multiplier RTL	Design 16-bit multiplier (Booth recommended)	Verified multiplier
4	Accumulator RTL	32-bit accumulator + register	Verified accumulator
5	Pipelined MAC	Integrate multiplier + accumulator with pipeline	MAC RTL
6	Convolution Engine	Design simple 3×3 convolution using MAC	CNN core RTL
7	Simulation	Full testbench for convolution	Verified CNN core
8	FPGA Synthesis	Run synthesis & implementation	LUT, DSP, timing report

PHASE 2 — ASIC IMPLEMENTATION (Cadence)

Tools:
- Xcelium
- Genus
- Innovus
- Tempus

Week	Stage	Work to Do	Output
9	RTL Optimization	Remove FPGA primitives, ASIC-ready RTL	Clean RTL
10	Synthesis (Genus)	Apply constraints, generate netlist	Area, timing, power report
11	Floorplanning	Import netlist, define core area	Floorplan snapshot
12	Placement + CTS	Place cells, clock tree synthesis	Placement & skew report
13	Routing + GDS-II	Complete routing, DRC check	Final layout + GDS-II
14	Post-Layout STA	Extract parasitics, compare PPA	Final PPA table

Architecture	Area	Delay	Power
FPGA MAC	LUT/DSP	Fmax	Est.
ASIC MAC	μm^2	ns	mW

Optional:

- Non-pipelined vs pipelined
- 16-bit vs 8-bit

🔥 If You Want More Research Strength

Add ONE of these:

- ✅ Clock gating for low power
- ✅ Systolic 2×2 MAC array
- ✅ Approximate multiplier
- ✅ Power gating

Architecture	Area	Delay	Power	Throughput
Booth MAC				
Pipelined Booth MAC				
2×2 Systolic MAC				

TOOLS USED

Tool	Purpose
—	Architecture theory
Xcelium + SimVision	RTL simulation of adders
Xcelium	ALU functional verification
Genus	Logic synthesis
Xcelium	Gate-level simulation
Innovus (demo)	PD flow introduction
Innovus	Floorplanning
Innovus	Placement
Innovus	Clock Tree Synthesis
Innovus	Routing + GDS-II generation
Tempus / Innovus STA	Post-layout timing analysis
Excel / MATLAB / Python	PPA graph comparison & report